# AC2204 Marking Scheme

## Introduction to Management Accounting · Winter 2020 · Paper 1

| | |
|---|---|
| **Examination Session and Year** | Winter 2020 |
| **Module Code** | AC2204 |
| **Module Title** | Introduction to Management Accounting |
| **Paper Number** | Paper 1 |
| **External Examiner** | Professor Ivo De Loo |
| **Head of the Department** | Professor Mark Mulcahy |
| **Internal Examiners** | Dr Margaret Healy |
| **Duration of Paper** | 1.5 hours |
| **Special Requirements** | Show your workings; marks will be awarded for workings. Separate answer books are not required. |

**Instructions to Candidates (as per exam paper):** Answer three questions in total. Attempt Question One from Section A (compulsory) and TWO questions from Section B.

**Own-figure rule:** where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

**Marking philosophy:** this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

### Question Overview

| Question | Scenario | Topics | Marks |
|---|---|---|---|
| Q1 (compulsory) | Applet Ltd | Marginal vs absorption costing across two consecutive years (opening and closing stock carried forward), income statement format, profit reconciliation | 40 |
| Q2 (Section B option) | Olaf Ltd | CVP analysis, breakeven, margin of safety, operating leverage, methods for reducing subjectivity in cost estimation | 30 |
| Q3 (Section B option) | Echo Ltd | Traditional (absorption) costing vs activity-based costing, product cost vs period cost, how ABC treats costs differently | 30 |
| Q4 (Section B option) | Statement evaluation | Breakeven/MOS/operating leverage verification, fixed cost per unit, capital vs labour intensity, high-low method, opportunity vs differential costs | 30 |

QUESTION 1

# Applet Ltd

Total: 40 marks

Applet Ltd makes and sells a single product, with data given for its first two years of operation (year ended 31/08/2019 and year ended 31/08/2020). Unlike a single-year scenario, the 300 units of closing stock from 2019 carry forward as opening stock into 2020, valued at the *2019* cost rates. Candidates are required to (A) calculate unit cost and operating profit for both years under both costing methods, (B) explain why the income statement format differs between the two methods, and (C) reconcile the profit figures for both years.

| Part | Requirement | Marks |
|---|---|---|
| (A) | Unit cost and operating profit for both years – marginal and absorption costing | 25 |
| (B) | Explain why the income statement format differs between the two methods | 5 |
| (C) | Profit reconciliation for both years | 10 |

**Question 1(A)** **25 marks**

Applet Ltd, first two years of operation. Year ended 31/08/2019: Sales 2,000 units; Selling price €22.50; Production 2,300 units; Variable manufacturing €24,840; Fixed manufacturing €12,190; Variable marketing €4,000; Fixed marketing €5,000. Year ended 31/08/2020: Sales 2,600 units; Selling price €23.00; Production 2,400 units; Variable manufacturing €28,800; Fixed manufacturing €13,200; Variable marketing €5,300; Fixed marketing €5,000. (A) Calculate, for each of the two years, the unit cost and the operating profit using (i) Marginal costing and (ii) Absorption costing.

## Full Model Answer – Year ended 31/08/2019

### Preliminary workings, 2019

This is Applet Ltd's first year, so there is no opening stock. Closing stock = Production − Sales = 2,300 − 2,000 = 300 units. Variable manufacturing cost per unit (2019) = €24,840 ÷ 2,300 = €10.80. Fixed manufacturing overhead per unit (2019, absorption costing only) = €12,190 ÷ 2,300 = €5.30.

### Unit costs, 2019

| Item | € |
|---|---|
| Marginal costing unit cost (variable manufacturing only) | 10.80 |
| Absorption costing unit cost (€10.80 + €5.30 fixed) | 16.10 |

### Income statement, year ended 31/08/2019 – marginal costing

| Item | € |
|---|---|
| Sales revenue (2,000 units × €22.50) | 45,000 |
| Less: Variable cost of goods sold (2,000 units × €10.80) | (21,600) |
| Less: Variable marketing costs | (4,000) |
| **Contribution margin** | **19,400** |
| Less: Fixed manufacturing costs | (12,190) |
| Less: Fixed marketing costs | (5,000) |
| **Operating profit (marginal costing), 2019** | **2,210** |

### Income statement, year ended 31/08/2019 – absorption costing

| Item | € |
|---|---|
| Sales revenue (2,000 units × €22.50) | 45,000 |
| Less: Cost of goods sold (2,000 units × €16.10) | (32,200) |
| **Gross profit** | **12,800** |
| Less: Variable marketing costs | (4,000) |
| Less: Fixed marketing costs | (5,000) |
| **Operating profit (absorption costing), 2019** | **3,800** |

**Closing stock carried forward into 2020:** 300 units of closing stock at 31/08/2019 are valued at €3,240 under marginal costing (300 × €10.80) and €4,830 under absorption costing (300 × €16.10). These become the *opening* stock values for 2020, carried at 2019's cost rates — not recalculated using 2020's rates.

## Full Model Answer – Year ended 31/08/2020

### Preliminary workings, 2020

Opening stock = 300 units (carried forward from 2019, at 2019 values: €3,240 marginal / €4,830 absorption). Closing stock = Opening stock + Production − Sales = 300 + 2,400 − 2,600 = 100 units. Variable manufacturing cost per unit (2020) = €28,800 ÷ 2,400 = €12.00. Fixed manufacturing overhead per unit (2020, absorption costing only) = €13,200 ÷ 2,400 = €5.50.

### Unit costs, 2020

| Item | € |
|---|---|
| Marginal costing unit cost (variable manufacturing only) | 12.00 |
| Absorption costing unit cost (€12.00 + €5.50 fixed) | 17.50 |

### Income statement, year ended 31/08/2020 – marginal costing

| Item | € |
|---|---|
| Sales revenue (2,600 units × €23.00) | 59,800 |
| Opening stock (300 units, at 2019 marginal value) | 3,240 |
| Add: Variable cost of production (2,400 units × €12.00) | 28,800 |
| Less: Closing stock (100 units × €12.00) | (1,200) |
| Variable cost of goods sold | 30,840 |
| Less: Variable marketing costs | (5,300) |
| **Contribution margin** | **23,660** |
| Less: Fixed manufacturing costs | (13,200) |
| Less: Fixed marketing costs | (5,000) |
| **Operating profit (marginal costing), 2020** | **5,460** |

### Income statement, year ended 31/08/2020 – absorption costing

| Item | € |
|---|---|
| Sales revenue (2,600 units × €23.00) | 59,800 |
| Opening stock (300 units, at 2019 absorption value) | 4,830 |
| Add: Full production cost (2,400 units × €17.50) | 42,000 |
| Less: Closing stock (100 units × €17.50) | (1,750) |
| Cost of goods sold | 45,080 |
| **Gross profit** | **14,720** |
| Less: Variable marketing costs | (5,300) |
| Less: Fixed marketing costs | (5,000) |
| **Operating profit (absorption costing), 2020** | **4,420** |

**Notable result:** Unlike 2019 (where absorption profit exceeded marginal profit), in 2020 marginal costing profit (€5,460) exceeds absorption costing profit (€4,420). This is because sales (2,600 units) exceeded production (2,400 units) in 2020, meaning stock levels fell during the year — more fixed overhead was released from opening stock than was deferred into closing stock. This reversal is explored fully in part (C).

## Mark Allocation

| Marks | Criteria |
| --- | --- |
| **22–25** | Both years' unit costs correct; all four income statements (2019 marginal/absorption, 2020 marginal/absorption) correctly prepared, with 2020 correctly incorporating the opening stock carried forward at 2019 values. |
| **16–21** | Substantially correct with a minor error, most commonly in correctly carrying forward and valuing the 2020 opening stock. |
| **9–15** | 2019 figures correct; 2020 figures show a fundamental misunderstanding of how opening stock should be valued (e.g. incorrectly revalued at 2020 rates), or other significant errors. |
| **0–8** | Limited understanding of the distinction between the two costing methods; minimal correct workings. |

**Question 1(B)** **5 marks**

Explain why the format of the income statement differs under each method of costing.

## Full Model Answer

### Marginal costing format – classification by cost behaviour

The marginal costing income statement is structured around cost *behaviour*: all variable costs (both manufacturing and marketing) are deducted from sales to arrive at a contribution margin subtotal, before fixed costs (again, both manufacturing and marketing) are deducted in total to arrive at operating profit. This format directly supports CVP analysis and short-term decision-making, since the contribution margin is a key input to breakeven and what-if analysis.

### Absorption costing format – classification by function

The absorption costing income statement is structured around cost *function*: all production costs (variable and fixed manufacturing combined) are deducted from sales to arrive at a gross profit subtotal, before non-production costs (variable and fixed marketing/selling combined) are deducted to arrive at operating profit. This format follows the conventional structure required for external financial reporting under IAS 2, distinguishing cost of sales from operating expenses rather than distinguishing costs by how they behave with volume.

**Key point:** The underlying difference in format reflects two different classification bases: marginal costing classifies every cost as fixed or variable; absorption costing classifies every cost as production or non-production. This is why marginal costing's key subtotal is 'contribution margin' while absorption costing's key subtotal is 'gross profit' — they are answering different questions.

## Mark Allocation

| Marks | Criteria |
|---|---|
| **4–5** | Clear explanation that the two formats reflect different underlying cost classification bases (behaviour vs function), correctly identifying the contribution margin and gross profit subtotals as the key structural difference. |
| **3** | Reasonable explanation of the different subtotals used, without fully explaining the underlying classification basis. |
| **1–2** | Some relevant point made but explanation is vague or incomplete. |
| **0** | No meaningful explanation provided. |

## Question 1(C)

10 marks

Prepare, for each of the two years, a reconciliation which explains the difference in the operating profits resulting from the use of the two costing methods.

## Full Model Answer

### Profit reconciliation, year ended 31/08/2019

| Item | € |
|---|---|
| Operating profit – marginal costing, 2019 | 2,210 |
| Add: Fixed manufacturing overhead carried forward in closing stock (300 units × €5.30) | 1,590 |
| Less: Fixed manufacturing overhead carried forward in opening stock (none – first year of trading) | 0 |
| **Operating profit – absorption costing, 2019** | **3,800** |

### Profit reconciliation, year ended 31/08/2020

| Item | € |
|---|---|
| Operating profit – marginal costing, 2020 | 5,460 |
| Add: Fixed manufacturing overhead carried forward in closing stock (100 units × €5.50) | 550 |
| Less: Fixed manufacturing overhead carried forward in opening stock (300 units × €5.30, the 2019 rate) | (1,590) |
| **Operating profit – absorption costing, 2020** | **4,420** |

**Why the direction reverses between the two years:** In 2019, production (2,300) exceeded sales (2,000), so stock levels increased, and absorption costing profit exceeded marginal costing profit. In 2020, sales (2,600) exceeded production (2,400), so stock levels fell (from 300 to 100 units), and more fixed overhead was released from opening stock than was deferred into closing stock — so marginal costing profit exceeded absorption costing profit for 2020. This illustrates the general rule: absorption costing profit exceeds marginal costing profit whenever stock levels rise during the period, and falls below it whenever stock levels fall.

## Mark Allocation

| Marks | Criteria |
|---|---|
| 8–10 | Both years' reconciliations correct, including the 2020 opening stock adjustment using the correct (2019) fixed overhead rate; the reversal in direction between the two years correctly explained by reference to the relationship between production and sales. |
| 6–7 | Both reconciliations substantially correct with a minor error, most likely in the 2020 opening stock figure. |
| 3–5 | 2019 reconciliation correct; 2020 reconciliation shows a fundamental misunderstanding of the opening stock adjustment or its rate. |
| 0–2 | No meaningful reconciliation provided for either year. |

**QUESTION 1 TOTAL — 40 MARKS**

QUESTION 2

# Olaf Ltd

Total: 30 marks

Olaf Ltd manufactures wooden tree houses for children under the single model 'Crann'. Candidates are required to (A) calculate and interpret five core CVP metrics, and (B) address management's concern about subjectivity in cost estimation by outlining two more objective estimation methods.

| Part | Requirement | Marks |
|---|---|---|
| (A) | Breakeven (units and revenue), operating profit, margin of safety, operating leverage – with explanation of usefulness | 20 |
| (B) | Two methods to reduce subjectivity in estimating future costs | 10 |

## Question 2(A)

**20 marks**

Olaf Ltd, 'Crann' tree house: Unit sales price €1,200; Variable costs per unit €720; Total fixed expenses €120,000. Olaf Ltd plan to sell 300 tree houses. Calculate each of the following and briefly explain their usefulness for financial planning: (i) Breakeven point in units; (ii) Breakeven point in sales revenue, using the contribution margin ratio; (iii) Operating profit or loss; (iv) Percentage margin of safety; (v) Degree of operating leverage.

### Full Model Answer

Contribution margin per unit = €1,200 − €720 = €480. Contribution margin ratio = €480 ÷ €1,200 = 40%.

#### (i) Breakeven point in units – 250 tree houses

Breakeven units = Fixed costs ÷ Contribution margin per unit = €120,000 ÷ €480 = **250 units**. *Usefulness:* shows Olaf Ltd the minimum number of tree houses that must be sold before the business breaks even, useful for setting a minimum sales target and assessing how much cushion the 300-unit sales plan provides.

#### (ii) Breakeven point in sales revenue – €300,000

Breakeven revenue = Fixed costs ÷ Contribution margin ratio = €120,000 ÷ 40% = **€300,000**. *Usefulness:* expresses the breakeven point in euro sales terms, useful for setting revenue targets and comparing directly against budgeted sales revenue.

#### (iii) Operating profit – €24,000

Operating profit = (300 × €480) − €120,000 = €144,000 − €120,000 = **€24,000**. *Usefulness:* shows the expected bottom-line profit at the planned sales volume of 300 units, forming the basis for budgeting and assessing whether the plan is worthwhile.

#### (iv) Percentage margin of safety – 16.67%

Margin of safety % = (Planned sales − Breakeven sales) ÷ Planned sales = (300 − 250) ÷ 300 = 50 ÷ 300 = **16.67%**. *Usefulness:* indicates how far sales can fall from the planned 300 units before the company starts making a loss, giving management a measure of the risk built into the sales plan — a relatively low margin of safety here signals limited room for a sales shortfall.

#### (v) Degree of operating leverage – 6.0

Degree of operating leverage = Total contribution margin ÷ Operating profit = €144,000 ÷ €24,000 = **6.0**. *Usefulness:* indicates that a 1% increase in sales revenue would increase operating profit by approximately 6%, highlighting that profit is highly sensitive to changes in sales volume at this level of activity, consistent with the relatively low margin of safety identified above.

### Mark Allocation

| Marks | Criteria |
|---|---|
| 18–20 | All five metrics correctly calculated, with a clear and specific explanation of the usefulness of each for financial planning (not generic statements). |
| 13–17 | Four or five metrics correctly calculated with reasonable explanations; one weaker or missing explanation. |
| 8–12 | Most metrics correctly calculated but explanations are generic or missing; or a systematic error affects several figures. |
| 0–7 | Limited understanding of CVP metrics; minimal correct workings. |

## Question 2(B)

**10 marks**

Management at Olaf Ltd are concerned about the level of subjectivity there appears to be in estimating the future costs of the organisation. Prepare a short note outlining two possible methods that could be used to address their concerns.

## Full Model Answer

### 1. Regression analysis (least squares method)

Regression analysis fits a straight line mathematically through all available historical cost/activity data points, minimising the sum of the squared vertical distances between the line and each actual observation. This produces an objective, mathematically determined estimate of both the fixed cost (the regression intercept) and the variable cost per unit (the regression slope), removing the subjective judgement involved in visually fitting a line (as in the scattergraph method) or in selecting just two data points (as in the high-low method, which can be badly distorted by an unrepresentative high or low observation). Statistical measures such as the correlation coefficient ($R^2$) can additionally be used to assess how well the resulting cost function actually fits the underlying data, giving management an objective measure of confidence in the estimate.

### 2. Engineering (industrial engineering) method

The engineering method estimates costs from first principles by analysing the physical inputs (materials, labour time, machine time) technically required to produce one unit, based on specifications, time-and-motion studies, or engineering standards, rather than by extrapolating from historical accounting data at all. Because it does not rely on interpreting past cost behaviour or fitting a line to historical data, it removes the subjectivity associated with all the other estimation methods, though it can be time-consuming and costly to carry out and depends on the reliability of the technical/engineering analysis performed.

**Why these two methods reduce subjectivity:** The high-low method (using only two data points) and the scattergraph method (requiring a hand-drawn 'best fit' line) both involve a meaningful degree of subjective judgement in their application. Regression analysis replaces that judgement with an objective statistical calculation, while the engineering method sidesteps historical cost data analysis altogether in favour of a technical, first-principles approach.

## Mark Allocation

| Marks | Criteria |
|---|---|
| **8–10** | Two distinct, well-explained methods given, each clearly linked to how it reduces the subjectivity present in simpler cost estimation techniques (high-low, scattergraph). |
| **5–7** | Two methods identified with reasonable explanation; the link to reducing subjectivity specifically not always made explicit. |
| **2–4** | One method identified with limited explanation. |
| **0–1** | No relevant methods identified. |

**QUESTION 2 TOTAL — 30 MARKS**

QUESTION 3

# Echo Ltd

Total: 30 marks

Echo Ltd currently uses traditional (absorption) costing, absorbing production overheads of €122,400 on the basis of machine hours across three products. The Director of Manufacturing has questioned the value of activity-based costing, arguing that knowing whether a cost is a product cost or a period cost should be enough. Candidates are required to (A) recalculate product costs under both traditional and activity-based costing, and (B) respond to the Director's view.

| Part | Requirement | Marks |
|---|---|---|
| (A) | Total costs and cost per unit – traditional costing and activity-based costing | 22 |
| (B) | Response to the Director's view; explain how ABC treats costs differently | 8 |

**Question 3(A)** **22 marks**

Echo Ltd absorbs production overheads (€122,400) on the basis of machine hours, at €24 per machine hour. Direct labour costs €30 per hour. Product A1: 200 units, €48 DM/unit, 2 MH/unit, 400 total MH, 4 DLH/unit. Product B2: 700 units, €23 DM/unit, 5 MH/unit, 3,500 total MH, 2 DLH/unit. Product D4: 300 units, €36 DM/unit, 4 MH/unit, 1,200 total MH, 1 DLH/unit. Cost pools: Quality 30% (units), Materials handling 25% (set-ups), Purchasing 15% (purchase orders), Maintenance 30% (machine hours). Purchase orders: A1=10, B2=6, D4=4 (total 20). Set-ups: A1=10, B2=7, D4=3 (total 20). (A) Calculate the total costs and cost per unit for each product using (i) Traditional (absorption) costing and (ii) Activity-based costing.

## Full Model Answer – (i) Traditional (absorption) costing

Overhead absorption rate = €122,400 ÷ 5,100 total machine hours (400 + 3,500 + 1,200) = €24 per machine hour, as given.

### Traditional (absorption) costing – cost per unit

| Item | Product A1 | Product B2 | Product D4 |
|---|---|---|---|
| Direct materials | 48.00 | 23.00 | 36.00 |
| Direct labour (DLH × €30) | 120.00 | 60.00 | 30.00 |
| Overhead (MH/unit × €24) | 48.00 | 120.00 | 96.00 |
| **Total cost per unit** | **216.00** | **203.00** | **162.00** |

### Traditional (absorption) costing – total costs

| Item | Product A1 (€) | Product B2 (€) | Product D4 (€) |
|---|---|---|---|
| Units produced | 200 | 700 | 300 |
| **Total cost (unit cost × units)** | **43,200** | **142,100** | **48,600** |

**Check:** Total overhead absorbed = €9,600 (A1) + €84,000 (B2) + €28,800 (D4) = €122,400, agreeing with the total production overhead given.

## Full Model Answer – (ii) Activity-based costing

### Working – cost pools and activity rates

| Cost pool | Pool amount (€) | Cost driver | Total activity | Rate |
|---|---|---|---|---|
| Quality (30%) | 36,720 | Units of output | 1,200 units | €30.60 per unit |
| Materials handling (25%) | 30,600 | No. of set-ups | 20 set-ups | €1,530.00 per set-up |
| Purchasing (15%) | 18,360 | No. of purchase orders | 20 orders | €918.00 per order |
| Maintenance (30%) | 36,720 | Machine hours | 5,100 MH | €7.20 per MH |
| **Total** | **122,400** | | | |

### Working – overhead allocated to each product

| Cost pool | Product A1 (€) | Product B2 (€) | Product D4 (€) |
|---|---|---|---|
| Quality (units × €30.60) | 6,120 | 21,420 | 9,180 |
| Materials handling (set-ups × €1,530) | 15,300 | 10,710 | 4,590 |
| Purchasing (orders × €918) | 9,180 | 5,508 | 3,672 |
| Maintenance (total MH × €7.20) | 2,880 | 25,200 | 8,640 |
| **Total overhead** | **33,480** | **62,838** | **26,082** |

**Check:** Total overhead allocated = €33,480 + €62,838 + €26,082 = €122,400, agreeing with the total production overhead given.

### Activity-based costing – total costs and cost per unit

| Item | Product A1 | Product B2 | Product D4 |
|---|---|---|---|
| Direct materials (total) | 9,600 | 16,100 | 10,800 |
| Direct labour (total) | 24,000 | 42,000 | 9,000 |
| Overhead (from above) | 33,480 | 62,838 | 26,082 |
| **Total cost** | **67,080** | **120,938** | **45,882** |
| Units produced | 200 | 700 | 300 |
| **Cost per unit** | **335.40** | **172.77** | **152.94** |

**Comparison:** ABC dramatically increases Product A1's unit cost (€216.00 → €335.40, a 55% increase), reduces Product B2's slightly (€203.00 → €172.77), and reduces Product D4's slightly (€162.00 → €152.94). Product A1, despite its low volume (200 units) and modest machine-hour usage, consumes a disproportionate share of set-ups (10 of the 20 total, i.e. one set-up for every 20 units produced) and purchase orders relative to its size, which the traditional machine-hour-based method entirely fails to capture.

## Mark Allocation

| Marks | Criteria |
|---|---|
| **18–22** | Traditional costing correctly calculated; all four ABC cost pool rates correctly calculated; overhead correctly allocated to each product; ABC total cost and cost per unit correct for all three products. |
| **13–17** | Substantially correct with minor computational errors in one or two pool rates or allocations. |
| **7–12** | Traditional costing correct; ABC approach understood but rates or allocations contain multiple errors. |
| **0–6** | Limited understanding of either costing method; minimal correct workings. |

**Question 3(B)** **8 marks**

The Director of Manufacturing stated: "I don't know why there is such a fuss about activity-based costing. A cost is a cost and surely it's enough to know whether it's a product cost or a period cost." Prepare a short note commenting on this view. In your note, explain how costs are considered in an activity-based costing system.

## Full Model Answer

**DISAGREE**

The Director's view conflates two entirely different, unrelated cost classification questions. Knowing that a cost is a product cost (as opposed to a period cost) says nothing about how accurately that cost has been assigned to individual products — and this second question is exactly the one ABC addresses.

### Product cost vs period cost is a different question from allocation accuracy

The product cost/period cost distinction is about *when* a cost is expensed: product costs (direct materials, direct labour, manufacturing overhead) are inventoried and expensed only when the related unit is sold, while period costs (e.g. selling and administrative expenses) are expensed immediately, regardless of production or sales volume. This classification applies equally under both traditional absorption costing and ABC — ABC does not create a new category of cost sitting alongside product and period cost. What ABC changes is entirely different: *how* manufacturing overhead (a product cost under either system) is traced to individual products.

### How costs are considered differently in an ABC system

Under the traditional system used by Echo Ltd, all €122,400 of manufacturing overhead is a 'product cost' and is absorbed into products using a single, volume-based driver (machine hours), regardless of what actually causes each portion of that overhead to be incurred. ABC instead recognises that overhead costs are caused by different types of activity — some driven by units produced (quality), some by number of production batches or set-ups (materials handling), some by number of purchase orders (purchasing), and some by machine hours (maintenance) — and traces each cost pool to products using the driver that actually causes it. As shown in part (A), this reveals that Product A1 was substantially undercosted under the traditional system (€216.00 vs a true ABC cost of €335.40) because its low volume disguised a disproportionately high consumption of set-up and purchase-order activity.

**Conclusion for the Director:** Simply knowing that manufacturing overhead is a 'product cost' tells Echo Ltd nothing about whether that €122,400 has been fairly and accurately shared out between Product A1, B2, and D4. ABC is not a competing classification to product/period cost; it is a more refined method of allocating the product cost category specifically, which matters directly for pricing, product mix, and profitability decisions.

## Mark Allocation

| Marks | Criteria |
|---|---|
| 7–8 | Correct identification that the Director conflates two unrelated classification questions; clear explanation that ABC operates within the 'product cost' category and refines how overhead is traced to products using multiple activity-based drivers rather than a single volume-based driver; reference to the specific distortion revealed in part (A). |
| 5–6 | Good response identifying the conflation with reasonable explanation of how ABC treats costs differently; limited or no reference to part (A)'s figures. |
| 2–4 | Some relevant discussion but generic, or a one-sided response that does not clearly explain how ABC's treatment of overhead differs from the traditional approach. |
| 0–1 | Agrees with the Director's view without meaningful justification, or no meaningful discussion provided. |

**QUESTION 3 TOTAL — 30 MARKS**

QUESTION 4

# Statement Evaluation

Total: 30 marks

Five independent statements covering breakeven/margin of safety/operating leverage verification, fixed cost per unit behaviour, operating leverage in different cost structures, the high-low method, and opportunity vs differential costs. For each statement, candidates must state whether they agree or disagree, with supporting calculations and explanation.

| Part | Topic | Marks |
|---|---|---|
| (A) | Decimal Ltd – breakeven, margin of safety and operating leverage verification | 6 |
| (B) | Fixed cost per unit as production increases | 6 |
| (C) | Operating leverage: capital vs labour intensive | 6 |
| (D) | Moto Ltd – high-low method cost estimate | 6 |
| (E) | Opportunity costs vs differential costs | 6 |

## Question 4(A)

6 marks

Decimal Ltd produces a single product. If total variable expenses are €73,000, total fixed costs are €27,930, total sales are 20,000 units, totalling €122,000; then breakeven point is 11,400 units, the margin of safety is 43% and operating leverage is 4.6 times.

## Full Model Answer

PARTIALLY CORRECT

The break-even point and margin of safety figures given in the statement are correct, but the operating leverage figure of 4.6 times is incorrect — the correct figure is approximately 2.33. Overall, the statement should be assessed as **disagree**, since not all three figures asserted are correct.

### Working

| Item | Value |
|---|---|
| Selling price per unit (€122,000 ÷ 20,000) | €6.10 |
| Variable cost per unit (€73,000 ÷ 20,000) | €3.65 |
| Contribution margin per unit | €2.45 |
| **Break-even point (€27,930 ÷ €2.45)** | **11,400 units – correct as stated** |
| **Margin of safety ((20,000 − 11,400) ÷ 20,000)** | **43% – correct as stated** |
| Total contribution margin (20,000 × €2.45) | €49,000 |
| Operating profit (€49,000 − €27,930) | €21,070 |
| **Operating leverage (€49,000 ÷ €21,070)** | **2.33 – NOT 4.6 as stated** |

## Mark Allocation

| Marks | Criteria |
|---|---|
| **5–6** | All three figures independently recalculated and verified; correct verdict of disagree, with the error in the operating leverage figure specifically identified and the correct value (2.33) given. |
| **3–4** | Break-even point and margin of safety correctly verified; operating leverage recalculated but the final verdict not clearly stated, or a minor computational error. |
| **1–2** | Some correct workings but the operating leverage error not identified. |
| **0** | No meaningful attempt to verify the figures. |

## Question 4(B)

6 marks

Fixed costs per unit remain constant as the level of activity increases within the relevant range.

## Full Model Answer

**DISAGREE**

It is **total** fixed costs that remain constant within the relevant range, not fixed cost **per unit**. Fixed cost per unit actually decreases as activity increases.

> **Why fixed cost per unit falls, not stays constant**
> Fixed cost per unit is calculated as total fixed cost divided by the number of units of activity. Since total fixed cost is unchanged within the relevant range, as the number of units increases, that same total is spread over more units, causing the fixed cost per unit to fall. For example, from Question 2, Olaf Ltd's €120,000 total fixed costs give a fixed cost per unit of €400 at 300 units produced; at 400 units, the same €120,000 total would give a fixed cost per unit of only €300.

### Mark Allocation

| Marks | Criteria |
|---|---|
| **5–6** | Correct disagreement with clear explanation of why fixed cost per unit falls (not remains constant) as volume rises, correctly distinguishing total fixed cost (which does remain constant) from fixed cost per unit, with an illustrative example. |
| **3–4** | Correct disagreement with reasonable explanation; illustrative example missing. |
| **1–2** | Correct disagreement but with minimal or unclear explanation. |
| **0** | Incorrect verdict (agrees with the statement) or no meaningful explanation. |

**Question 4(C)** **6 marks**

Operating leverage at a given level of sales in a capital intensive company will tend to be lower than in a labour intensive company with the same total revenues and total expenses.

## Full Model Answer

**DISAGREE**

Operating leverage tends to be **higher**, not lower, in a capital-intensive company relative to a labour-intensive company with the same total revenues and total expenses.

> **Why capital intensity increases operating leverage**
> A capital-intensive company has a cost structure weighted more heavily towards fixed costs (e.g. depreciation on machinery) and correspondingly lower variable costs per unit, compared to a labour-intensive company with the same total revenue and total costs. Operating leverage is calculated as total contribution margin ÷ operating profit. Because the capital-intensive company's higher fixed cost proportion means a larger share of each sales euro is retained as contribution (to eventually cover those fixed costs), its total contribution margin is higher relative to operating profit at any given sales level, giving it a **higher**, not lower, degree of operating leverage — meaning its profit is more sensitive to changes in sales volume.

### Mark Allocation

| Marks | Criteria |
|---|---|
| **5–6** | Correct disagreement with a clear explanation of why a higher fixed cost proportion increases operating leverage, correctly identifying that the statement has the direction reversed. |
| **3–4** | Correct disagreement with reasonable explanation but limited detail on the underlying mechanism. |
| **1–2** | Correct verdict stated but with minimal or unclear explanation. |
| **0** | Incorrect verdict (agrees with the statement) or no meaningful explanation. |

## Question 4(D)

6 marks

Monthly activity levels and associated costs at Moto Ltd for the 2019 financial year: HIGH: May, 3,192 units, total cost €69,240. LOW: November, 1,963 units, total cost €44,660. If it is planned to produce 2,100 units in December 2020, then based on the cost function for 2019, the expected cost is €47,400.

## Full Model Answer

**AGREE**

The expected cost of €47,400 given in the statement is correct.

### Working – high-low method

| Item | Value |
|---|---|
| Variable cost per unit ((€69,240 − €44,660) ÷ (3,192 − 1,963)) | (€24,580 ÷ 1,229) = €20.00 per unit |
| Fixed cost (at high point: €69,240 − (3,192 × €20)) | €5,400 |
| Fixed cost (at low point: €44,660 − (1,963 × €20)) – check | €5,400 |
| Cost function | Total cost = €5,400 + (€20 × units) |

### Estimated cost at 2,100 units (December 2020)

| Item | Value |
|---|---|
| €5,400 + (€20 × 2,100) | €5,400 + €42,000 |
| **Estimated cost** | **€47,400** |

## Mark Allocation

| Marks | Criteria |
|---|---|
| **5–6** | Correct agreement; variable cost per unit and fixed cost correctly calculated and cross-checked at both points; correct cost function applied to confirm €47,400. |
| **3–4** | Correct agreement with a minor computational error, or the cross-check at the second point omitted. |
| **1–2** | High-low method attempted with the correct general approach but a significant computational error. |
| **0** | Incorrect verdict (disagrees with the statement) or no meaningful workings. |

## Question 4(E)

6 marks

Opportunity costs are costs that record the difference between two or more decision alternatives.

### Full Model Answer

DISAGREE

The description given in the statement is the definition of a **differential cost**, not an opportunity cost.

#### What an opportunity cost actually is

An opportunity cost is the potential benefit that is given up (forgone) when one course of action is chosen over the next best alternative. It does not appear anywhere in the accounting records, but is highly relevant to decision-making because it represents a genuine economic sacrifice.

#### What the statement is actually describing

The difference in cost between two or more decision alternatives is the definition of a **differential cost** (also called an incremental cost). While both concepts are relevant to decision-making, they measure different things: a differential cost is the change in *actual, recorded* cost between alternatives, whereas an opportunity cost is the value of a forgone *benefit* that is never recorded in the accounts at all.

### Mark Allocation

| Marks | Criteria |
|---|---|
| **5–6** | Correct disagreement with accurate definitions of both opportunity cost and differential cost, clearly identifying that the statement has the two terms swapped. |
| **3–4** | Correct disagreement with an accurate definition of one of the two terms. |
| **1–2** | Correct disagreement but with vague or partially inaccurate definitions. |
| **0** | Incorrect verdict (agrees with the statement) or no meaningful explanation. |

**QUESTION 4 TOTAL — 30 MARKS**